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## **Surrogate and Dimensionality Reduction-Based Coupled Optimization of Well Pattern and Fracture Network in Tight Oil Reservoirs**

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### **Abstract**

Tight oil reservoirs have become a critical replacement domain for enhancing unconventional oil and gas reserves and production worldwide. The coupled optimization of well patterns and fracture networks is essential for achieving efficient development in such reservoirs. To overcome the limitations of existing methods, including insufficient coverage of optimization parameters, single-objective formulations, and low computational efficiency, this study proposes a surrogate-assisted coupled optimization method for well patterns and fracture networks with dimensionality reduction. Firstly, a dynamically evolving well pattern unit model is constructed based on spatial topology analysis and geometric transformation theory, while incorporating safety constraints, to quantitatively characterize the geometric configuration of well and fracture networks. On this basis, a multi-objective optimization model is established, aiming to maximize fracture-controlled reserves and achieve balanced and effective reservoir utilization, thereby simultaneously considering reservoir recovery and economic benefits. A surrogate-assisted evolutionary optimization algorithm, SADE–Sammon, is introduced. By combining dimensionality reduction mapping and surrogate-assisted iteration, this method efficiently achieves rapid convergence and global optimization in high-dimensional problems. The proposed approach is applied to a representative tight oil reservoir in Block G of the Bohai Bay Basin, China. The results demonstrate that, compared with conventional algorithms, computational efficiency is significantly improved, with the calculation time reduced by 43.14%. The optimized number of wells stabilizes between 8 and 10, with fracture-controlled reserves of approximately  $2.0 \times 10^4 \text{ m}^3$ . The economic net present value (NPV) increases from 60 million CNY to 120 million CNY. The coverage of high oil saturation zones improves from 37% to 81%, while the average oil saturation decreases by 17.3%. Field applications based on the optimization results were further conducted. In the pilot

area, the average daily liquid production per well in the first year increased from 3.5 m<sup>3</sup>/d to 11.6 m<sup>3</sup>/d. The fracture-controlled reserves increased by 20.41%, and the predicted 10-year cumulative oil production per well increased by 9000 m<sup>3</sup>. This method effectively overcomes the limitations of traditional optimization techniques and achieves a global coupled optimization of development scale, reservoir utilization, and economic performance, providing robust technical support and theoretical guidance for the efficient development of tight oil reservoirs.

**Keywords:** Tight Oil Reservoirs, Hydraulic Fracturing Horizontal Wells, Well Pattern and Fracture Network, Coupled Optimization, SADE–Sammon Algorithm

## Introduction

Tight oil reservoirs, characterized by abundant global reserves, have become a critical replacement domain for boosting unconventional oil and gas production (Dorhjie et al., 2023; Liu, 2023). However, the development of these reservoirs is typically constrained by poor pore connectivity, extremely low permeability, and strong heterogeneity, which lead to limited reservoir utilization, rapid production decline, and low recovery efficiency (Wang et al., 2024a; Zhou et al., 2024). Currently, the "horizontal well + volume fracturing" technique has emerged as the core strategy for the efficient development of tight oil reservoirs. By constructing multi-scale, complex fracture networks through hydraulic fracturing, this approach effectively improves flow conditions, enhances single-well productivity, and expands reservoir utilization (Han et al., 2021; Wu et al., 2022; Zhao et al., 2021). Nevertheless, the overall development performance is not solely governed by the fracturing design of individual wells but is significantly affected by the coupling relationship between well pattern and fracture parameters. Specifically, well spacing within the well pattern determines the degree of inter-well interference and the balance of reserve utilization, while fracture parameters control the drainage radius and the flow capacity in the near-wellbore region (Al-Khdeewi et al., 2024; Luo et al., 2021; Xu et al., 2024). Therefore, constructing a reasonably matched well pattern–fracture network system is essential for enhancing the development performance of tight oil reservoirs.

Based on the above understanding, some researchers have conducted systematic studies focusing on well pattern design. He et al. (2010) determined the optimal well spacing intervals under different stress sensitivities by employing numerical reservoir simulation, considering both maximum economic benefits and ultimate recovery factors. Yu and Sepehrnoori (2014) quantified fracture parameters using production performance data and optimized well spacing in the Bakken tight oil reservoir by incorporating geomechanical effects. Zhan et al. (2022) conducted numerical reservoir simulation to optimize various combinations of well pattern configurations and well spacing for pressure-sensitive, low-permeability reservoirs under waterflooding in the Shengli Oilfield, China. The results indicated that, under specific injection intensities, the inverted five-spot well pattern achieved higher reservoir utilization and better development performance. Wu et al. (2025) developed an integrated well spacing optimization workflow for tight oil reservoirs, which incorporated geological modeling, geomechanical modeling, hydraulic fracturing simulation, and numerical reservoir simulation. By integrating production performance data with fracture monitoring data, the optimal well spacing of 250 m was identified in the target area.

After the well pattern configuration and well spacing range had been largely established, subsequent studies further focused on optimizing fracture parameters, primarily investigating the impacts of fracture half-length, fracture width, and fracture conductivity on single-well productivity. Xu et al. (2022) developed a fracture optimization model that considered stress sensitivity and pore pressure, and performed multi-objective optimization targeting fracture half-length and fracture conductivity, which significantly improved development performance. Jie et al. (2022) conducted numerical reservoir simulation to optimize fracture parameters and fracture arrangements for hydraulically fractured horizontal wells in tight oil reservoirs, aiming to maximize cumulative oil production. The results demonstrated that staggered

fracture arrangements, combined with reasonable well spacing, can effectively enhance the development performance of both single wells and multi-lateral horizontal wells. Zhang et al. (2022) established a fully coupled numerical reservoir simulation model integrating porous media deformation and fluid flow for multi-stage fractured horizontal wells in tight oil reservoirs. With the objective of improving production forecasting accuracy, they systematically analyzed the effects of fracture parameters, fluid–solid coupling, and reservoir property evolution on production performance. The study concluded that fracture conductivity, spacing, number, and length must be jointly optimized to achieve optimal development results.

However, traditional well pattern and fracture network design typically adopts stepwise optimization, which often neglects the spatial coupling and flow control interactions between wells and fractures. With the increasing demand for precision development, well pattern–fracture network coupled optimization has become an important research trend in tight oil reservoir development. Some researchers have predefined limited well pattern–fracture network configurations and conducted scheme selection through numerical reservoir simulation. Although these methods offer certain engineering applicability, they heavily rely on manual experience, cannot fully explore high-dimensional parameter spaces, and cannot guarantee globally optimal solutions (Jia, 2022; Liao et al., 2022). To improve optimization efficiency and solution quality, Ma et al. (2020) developed an automatic well pattern–fracture network coupled optimization model based on the Simultaneous Perturbation Stochastic Approximation (SPSA) algorithm, enabling the joint optimization of key parameters such as well spacing, fracture length, and fracture conductivity. Wang et al. (2024b) proposed a well pattern–fracture network coupled optimization method for complex offshore low-permeability reservoirs and validated its applicability and engineering potential in multi-parameter, highly coupled scenarios through case studies. Although recent studies have made progress in the coupled optimization of well pattern and fracture network parameters, existing methods still face challenges such as limited optimization variables, single-objective formulations, and low computational efficiency. Therefore, it is urgently necessary to develop coupled optimization methods with broad adaptability and rapid convergence capabilities for complex real-world reservoirs to provide reliable technical support for the efficient development of tight oil reservoirs.

To overcome the aforementioned challenges, this study proposes a well pattern–fracture network coupled optimization method that integrates parametric modeling, multi-objective optimization, and intelligent algorithms. Firstly, a dynamically evolving well pattern unit model is constructed based on spatial topology and geometric transformation theory, enabling the quantitative characterization of the geometric configuration of well and fracture networks. Secondly, a coupled optimization model is established with the objectives of maximizing fracture-controlled reserves—accounting for differences in fracture-stimulated reservoir areas—and achieving balanced and effective reservoir utilization. The SADE–Sammon optimization algorithm is introduced to efficiently solve the high-dimensional optimization problem. Finally, the proposed method is applied to a representative tight oil reservoir in Block G of the Bohai Bay Basin, China, to perform coupled optimization design and field validation.

## Method and Workflow

### Parameterization of Well Pattern and Fracture Network for Coupled Optimization

To accurately characterize the spatial topological relationship between the well pattern and fracture network, as well as the fluid flow behavior under their coupled effects, key geometric parameters are defined, including horizontal well trajectory coordinates, perforation positions, fracture half-lengths, and fracture conductivity. Based on these parameters, a geometric model of a single hydraulically fractured horizontal well is constructed. Geometric transformation parameters, such as rotation angle, shear angle, and translation distance, are introduced to enable modular replication of the single-well model. Through this process, a reservoir-scale coupled model of the well pattern and fracture network is established.

**Single-Well Model.** In the single-well scale modeling, a parameter system applicable to horizontally fractured wells for volume fracturing in tight oil reservoirs is established based on the distribution characteristics of horizontal well trajectories and fractures. The core control variables include: wellhead planar center coordinates  $(X_0, Y_0)$ , horizontal section length  $L_h$ , horizontal well azimuth  $\alpha$ , number of fractures  $N_f$ , angle between fractures and wellbore  $\beta$ , fracture half-length  $L_f$ , fracture width  $W$ , fracture conductivity  $F_f$ , and fracture height  $H$ . Among these parameters, the fracture height is typically assigned the value of the reservoir thickness, and the fracture orientation is generally aligned with the direction of the maximum horizontal principal stress. Figure 1 illustrates the spatial topological relationship between the perforation points along the horizontal well section and their corresponding fracture endpoints.

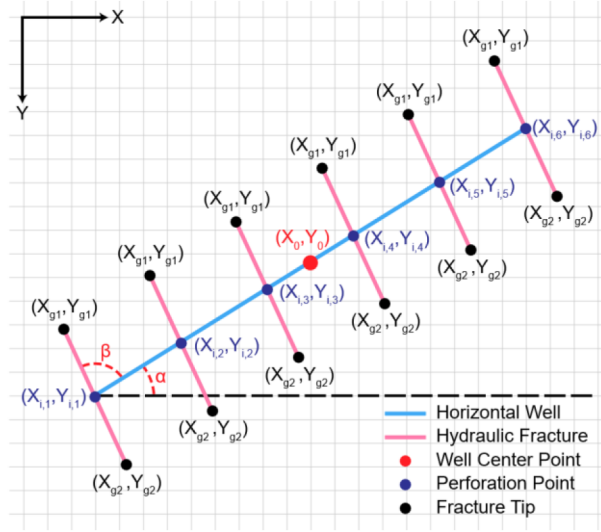


Figure 1—Schematic illustration of key parameter definitions and geometric relationships for a hydraulically fractured horizontal well. The figure shows the spatial distribution of perforation points along the wellbore and the orientation and extent of associated hydraulic fractures.

Based on the above fundamental parameters, the coordinates of the perforation positions and fracture endpoints can be calculated, enabling precise characterization of the geometric configuration of the fracture system at the single-well scale. In this study, an equally spaced perforation arrangement is adopted, and the calculation formulas for the perforation point coordinates are as follows:

$$\begin{aligned} X_{i,n} &= X_o + \left( \frac{L_h}{N_f - 1} \right) \cdot (n - 2) \cdot \cos \alpha \\ Y_{i,n} &= Y_o - \left( \frac{L_h}{N_f - 1} \right) \cdot (n - 2) \cdot \sin \alpha \end{aligned} \quad (1)$$

where  $n \in [2, N_f]$  is the perforation segment index, and  $i$  denotes the well index.

Based on the perforation point locations, the spatial coordinates of the two fracture endpoints on both sides can be further calculated. The coordinates of the first fracture endpoint are:

$$\begin{aligned} X_{g,1} &= X_{i,n} - L_f \cdot \cos(180^\circ - \alpha - \beta) \\ Y_{g,1} &= Y_{i,n} - L_f \cdot \sin(180^\circ - \alpha - \beta) \end{aligned} \quad (2)$$

The coordinates of the second fracture endpoint are:

$$\begin{aligned} X_{g,2} &= X_{i,n} - L_f \cdot \cos(180^\circ - \alpha - \beta) \\ Y_{g,2} &= Y_{i,n} - L_f \cdot \sin(180^\circ - \alpha - \beta) \end{aligned} \quad (3)$$

Based on the key parameter definitions and calculations above, a three-dimensional geometric model of the hydraulically fractured horizontal well at the single-well scale is constructed.

**Well Pattern Model.** Based on the three-dimensional geometric model of a single hydraulically fractured horizontal well, a deformable well pattern unit characterization model controlled by geometric parameters is established. The spatial configuration and evolution of the well pattern unit are quantitatively regulated by six geometric parameters: transverse scaling factor  $\Delta a$ , longitudinal scaling factor  $\Delta b$ , shear angle  $\gamma$ , rotation angle  $\theta$ , transverse translation distance  $\Delta x$ , and longitudinal translation distance  $\Delta y$ .

Specifically, the shear angle  $\gamma$  controls the degree of horizontal shearing within the unit, enabling topological transformation from a rectangle to a parallelogram. The scaling factors  $\Delta a$  and  $\Delta b$  adjust the transverse and longitudinal dimensions, thereby controlling the well pattern density. The translation distances  $\Delta x$  and  $\Delta y$  enable the overall displacement of the unit to match the boundaries of the effective reservoir development area. The rotation angle  $\theta$  adjusts the principal axis orientation of the well pattern to align with the direction of the reservoir's in-situ stress field. As shown in Figure 2, coordinated adjustment of these parameters enables dynamic evolution of the well pattern unit, supporting flexible optimization of the well pattern layout.

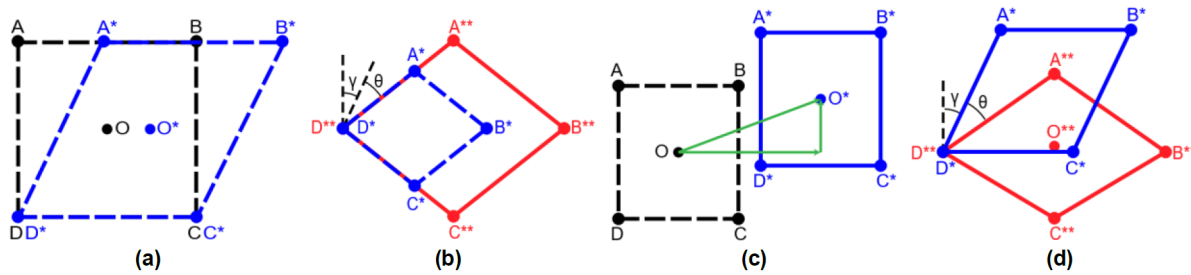


Figure 2—Schematic illustration of geometric transformations of a well pattern unit: (a) shear transformation, (b) scaling transformation, (c) translation transformation, and (d) rotation transformation.

During the optimization iteration process, the six geometric control parameters collectively form a unified two-dimensional affine transformation system. The final spatial position of any well point can be calculated from its initial reference coordinates through the following transformation relationship:

$$\begin{bmatrix} X' \\ Y' \end{bmatrix} = \mathbf{R}(\theta) \cdot \mathbf{S}(\Delta a, \Delta b, \gamma) \cdot \begin{bmatrix} X \\ Y \end{bmatrix} + \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix} \quad (4)$$

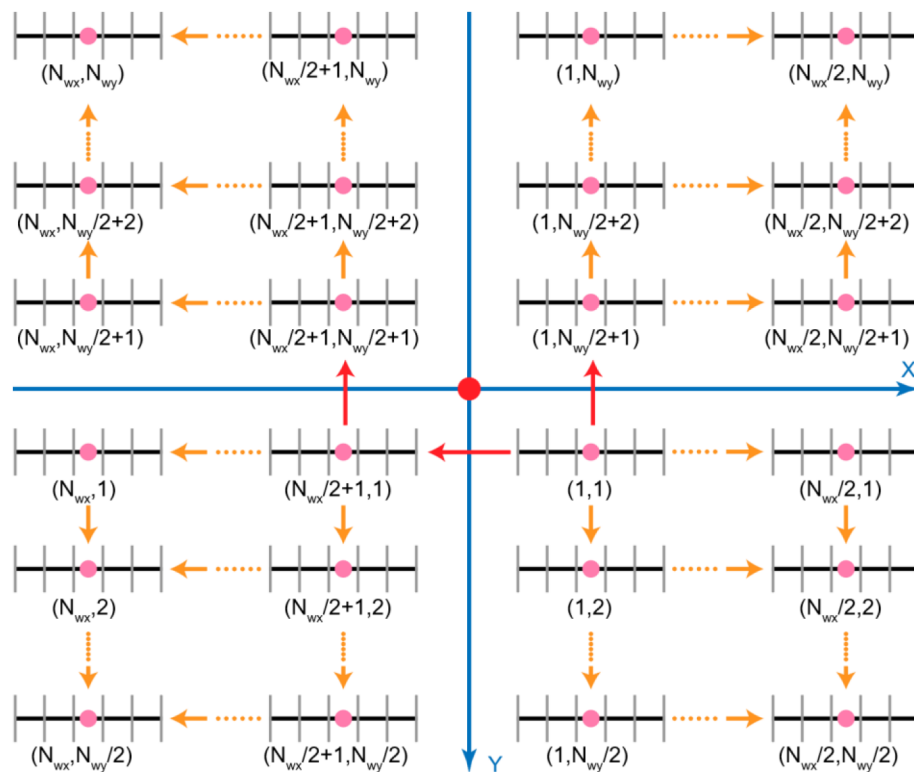
Where the rotation matrix  $\mathbf{R}(\theta)$  and the composite scaling-shear matrix  $\mathbf{S}(\Delta a, \Delta b, \gamma)$  are defined as follows:

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad (5)$$

$$\mathbf{S}(\Delta a, \Delta b, \gamma) = \begin{bmatrix} \Delta a & \Delta a \cdot \tan\gamma \\ 0 & \Delta b \end{bmatrix} \quad (6)$$

Based on the above transformation method, the overall well pattern structure is constructed on the basis of the single-well model through a recursive unit replication algorithm. As illustrated in Figure 3, starting from the geometric center of the reservoir, the affine-transformed well pattern units are iteratively tiled in all four quadrant directions. During the spatial expansion process, a ray casting detection algorithm is employed to continuously verify the geological validity of newly generated well points. Well points located within valid reservoir grids (with porosity greater than the threshold) are retained, while those falling outside the effective reservoir zones or beyond the boundary are automatically discarded. The iterative unit tiling and boundary detection procedure is repeatedly executed until the entire target area is fully covered. The final well pattern–fracture network is strictly coupled with the geological model.





**Figure 3—Schematic illustration of the well pattern construction process. The base well pattern unit is recursively tiled across four quadrants starting from the reservoir center to generate the full-field layout. Ray-tracing-based geometric constraints are applied to ensure that wells are placed only within valid reservoir zones.**

In addition, to ensure engineering feasibility and avoid issues such as well pattern overlap and fracture network intersection, a minimum inter-well distance constraint must be imposed. For any two horizontal wells, the minimum distance must satisfy the following condition:

$$\min_{i \neq j} |\mathbf{L}_i - \mathbf{L}_j| \geq d_{\min} \quad (7)$$

where  $d_{\min}$  is the predefined safety distance threshold.

To efficiently determine the minimum distance between well segments, a segment-to-segment minimum distance analytical algorithm is introduced. For any two well segments  $A$  and  $B$ , the segments are defined as finite straight lines  $L_a$  and  $L_b$ , with corresponding endpoints  $\mathbf{P}_1$ ,  $\mathbf{P}_2$  and  $\mathbf{Q}_1$ ,  $\mathbf{Q}_2$ . By parameterizing the segments, the coordinates of any points on  $L_a$  and  $L_b$  can be expressed as:

$$\mathbf{P}(s) = \mathbf{P}_1 + s(\mathbf{P}_2 - \mathbf{P}_1) \quad (8)$$

$$\mathbf{Q}(t) = \mathbf{Q}_1 + t(\mathbf{Q}_2 - \mathbf{Q}_1) \quad (9)$$

where  $s$  and  $t$  are scaling factors representing the relative positions of points along their respective segments, with  $s, t \in [0, 1]$ . When  $s = 0$ , the point is located at  $\mathbf{P}_1$ , and when  $s = 1$ , the point is located at  $\mathbf{P}_2$ . The physical meaning of parameter  $t$  is analogous.

The objective is to determine the minimum Euclidean distance between the two well segments, which is expressed as:

$$d_{ab} = \min_{s,t \in [0,1]} \|\mathbf{P}(s) - \mathbf{Q}(t)\| \quad (10)$$

An analytical solution for the above equation is derived using a closed-form method. The solution is obtained by constructing a distance function and solving the corresponding first-order derivative system, while incorporating boundary constraints. This approach directly yields the minimum distance with a computational complexity of  $O(1)$ , significantly improving the efficiency of the optimization process.

Based on the single-well and well pattern parameterization models described above, an intelligent modeling method for well pattern–fracture network construction in complex tight reservoirs is established. This method also integrates spatial safety distance constraints. The proposed modeling approach provides a solid foundation for the subsequent coupled optimization process.

### Coupled Optimization Mathematical Model

Based on the previously established well pattern–fracture network parameterization system, a coupled optimization mathematical model for tight oil reservoirs is developed. The model simultaneously considers the maximization of fracture-controlled reserves and the balanced, effective utilization of the reservoir. Fracture-controlled reserves characterize the effective stimulation capacity of the fracture system within the reservoir. This parameter is primarily determined by the spatial matching between the fracture distribution and reservoir properties. Balanced and effective reservoir utilization evaluates the synergy between the well pattern and the fracture system and reflects the uniformity of reservoir development. However, balanced utilization is difficult to quantify through a single physical indicator. Therefore, the economic net present value (NPV) is adopted in this study as a comprehensive evaluation metric.

**Objective Function.** To quantitatively evaluate the fracture-controlled reserves of each hydraulically fractured horizontal well, the previously established well pattern–fracture network geometric model is first coupled with the geological model of the target reservoir for numerical simulation. A ray-tracing algorithm is employed to identify the grid cells penetrated by fractures. Local grid refinement is then performed, and corrections are applied to fracture porosity, permeability, and conductivity parameters. Based on these adjustments, a multi-continuum reservoir numerical simulation model for the well-fracture system is established. On this basis, reservoir simulation is conducted to analyze the flow dynamics near the wellbore region and to delineate the actual drainage area of each well. Subsequently, the storage capacity of all grid cells within the drainage area is accumulated to calculate the fracture-controlled reserves  $Q_w$ , which is expressed as:

$$Q_w = \sum_{i=1}^{N_w} (A_i \cdot H_i \cdot \phi_i \cdot S_{o,i} / B_{o,i}) \quad (11)$$

where  $N_w$  is the number of horizontal wells,  $A_i$  is the drainage area of the  $i$ -th horizontal well,  $H_i$  is the average effective thickness within the drainage area,  $\phi_i$  is the average porosity within the drainage area,  $S_{o,i}$  is the average oil saturation within the drainage area, and  $B_{o,i}$  is the oil formation volume factor.

For horizontal wells developed through volume fracturing, the flow field within the well drainage area can be divided into four regions based on fluid flow characteristics: primary fracture zone, secondary fracture zone, matrix-stimulated zone, and unstimulated zone. The degree of reservoir stimulation varies significantly across these regions, leading to considerable differences in the recoverable reserves. To account for these differences, stimulation coefficients  $\delta$  are assigned to each region according to their respective flow characteristics and stimulation levels. The fracture-controlled reserves  $Q_f$  are then obtained by applying a weighted calculation of the well-controlled reserves and the corresponding stimulation coefficients. The calculation formula is as follows:

$$Q_f = \sum_{i=1}^{N_w} \delta_i \cdot Q_{w,i} \quad (12)$$

The economic net present value (NPV) is an important indicator in this study for evaluating balanced reservoir utilization and profitability. It comprehensively accounts for multiple factors, including oil and gas sales revenue, operating costs, capital investment, and the time value of money. The calculation formula is as follows:

$$NPV = \sum_{t=1}^T \frac{(R_t - C_t)}{(1+r)^t} - C_{init} \quad (13)$$

where  $R_t$  is the revenue in year  $t$ ,  $C_t$  is the operating expenditure in year  $t$ ,  $r$  is the discount rate,  $T$  is the total production period in years, and  $C_{init}$  is the initial capital investment.

where  $R_t$  is the sales revenue in year  $t$ ,  $C_t$  is the operating expenditure in year  $t$ ,  $r$  is the discount rate,  $T$  is the project life cycle (years), and  $C_{init}$  is the total initial investment.

The sales revenue  $R_t$  is calculated based on the cumulative oil production obtained from reservoir numerical simulation and the oil price. The operating expenditure  $C_t$  includes the comprehensive operating costs such as production operations, equipment maintenance, water injection operations, and water treatment. The initial investment  $C_{init}$  includes drilling, completion, hydraulic fracturing, and surface infrastructure costs. The total investment amount is dynamically adjusted according to the optimized well pattern scale and fracture parameters.

**Constraint Conditions.** The well pattern–fracture network coupled optimization problem addressed in this study can be formulated as a general mixed-constraint nonlinear multi-objective optimization model. The overall mathematical expression is as follows:

$$\max_{\mathbf{x} \in \chi} \mathbf{f}(\mathbf{x}) = [\mathbf{VF}(\mathbf{x}), \mathbf{NPV}(\mathbf{x})]^T \quad \text{s.t.} \quad \mathbf{g}(\mathbf{x}) \leq 0 \quad (14)$$

where  $\mathbf{f}(\mathbf{x})$  represents the dual-objective optimization function vector composed of fracture-controlled reserves and economic net present value.  $\chi$  represents the feasible domain defined by the boundary constraints of the control variables.  $\mathbf{g}(\mathbf{x}) \leq 0$  denotes the nonlinear inequality constraint set, which includes physical requirements such as safety distance and development feasibility.

To improve the convergence and stability of the model solution process, the control variables and constraint conditions are standardized. For physical variables with significant magnitude differences (such as fracture conductivity, fracture half-length, etc.), logarithmic transformation is applied to normalize them to a dimensionless interval of  $[0,1]$ . The normalization expression is as follows:

$$\tilde{x}_i = \frac{\log_{10}(x_i) - \log_{10}(x_i^{\min})}{\log_{10}(x_i^{\max}) - \log_{10}(x_i^{\min})}, \quad \tilde{x}_i \in [0, 1] \quad (15)$$

where  $x_i$  is the original optimization control variable,  $x_i^{\min}$  and  $x_i^{\max}$  are the predefined physical lower and upper bounds, and  $\tilde{x}_i$  is the normalized dimensionless variable after logarithmic transformation.

## Optimization Algorithm

The well pattern–fracture network coupled optimization model is characterized by high-dimensional variables and strong physical coupling. Additionally, each objective function evaluation requires invoking the reservoir numerical simulator, which leads to significant computational cost. Traditional optimization methods struggle to achieve efficient convergence under these conditions. To address this issue, this study introduces the SADE–Sammon algorithm (Chen et al., 2020), which integrates surrogate modeling with dimensionality reduction strategies. This approach enables efficient global solutions for high-dimensional production optimization problems within a limited number of reservoir simulation evaluations.

The overall optimization process adopts a dimensionality-reduction-assisted surrogate model iterative framework. This framework consists of four core steps: high-dimensional sampling, dimensionality reduction mapping, low-dimensional optimization, and global iterative updating. First, Latin Hypercube Design (LHD) is employed to perform initial sampling within the high-dimensional control parameter space. A training dataset is generated, and the objective function values are calculated through a limited number of reservoir simulations to establish the initial training database. Next, the Sammon mapping



algorithm is applied to perform nonlinear dimensionality reduction on the high-dimensional control parameters. The reduced parameters are projected into a low-dimensional space, effectively mitigating the "curse of dimensionality" commonly encountered in surrogate model construction. In the low-dimensional optimization phase, a Kriging surrogate model is used to fit the response relationship between the low-dimensional parameters and the objective function. The Differential Evolution (DE) algorithm is employed to rapidly iterate on the low-dimensional surrogate model. The Lower Confidence Bound (LCB) criterion is used to pre-screen potential high-quality solutions. These candidate solutions are then evaluated using the reservoir numerical simulator to obtain the true objective function values. Newly obtained high-quality samples are dynamically added to the training database. Through continuous iterative optimization, the dimensionality reduction mapping and the surrogate model are progressively updated, improving both optimization accuracy and convergence speed.

Through the collaborative iteration of dimensionality reduction and surrogate modeling, the SADE–Sammon algorithm significantly reduces the computational complexity of high-dimensional production optimization problems. It also maintains both global exploration capability and local exploitation accuracy. This method effectively improves the solution efficiency and result reliability of complex reservoir production optimization under limited computational resources.

## Case Study

### Overview of the Study Area

This study focuses on a typical tight oil reservoir located in Block G of the Bohai Bay Basin, China. The target area is situated within a structural depression belt, where the reservoir mainly consists of fan delta front sand bodies. The depositional facies exhibit typical thin interbedded sand and mud layers, with an average single-layer thickness of approximately 2.8 m. Vertically, the reservoir is divided into 18 oil-bearing sublayers, and the effective oil-bearing section per well generally ranges from 180 to 220 m. The reservoir is buried at considerable depth, with an initial formation pressure of 51.05 MPa, a saturation pressure of 25.27 MPa, and a formation temperature of 124 °C. The crude oil has a density of 0.75 g/cm<sup>3</sup> and a viscosity of 0.5 mPa·s, which is classified as a typical light oil tight reservoir.

The block has undergone three development stages: rolling development (January 2011 to December 2014), integrated fracturing and water injection development (January 2015 to December 2021), and volume fracturing horizontal well development (starting from 2022). Hydraulic fracturing has increased single-well productivity by 2.5 times and controlled geological reserves exceeding  $5 \times 10^4$  t, demonstrating a significant production enhancement. However, due to fracture closure, production from fractured wells declines rapidly, and the drainage in areas far from the wellbore remains insufficient. In addition, the injection–production well spacing is relatively large, and water injection after fracturing has not effectively expanded the well-controlled drainage area. As a result, part of the reserves remains in a low-efficiency or untapped state. Furthermore, the reservoir exhibits poor overall porosity and permeability, leading to a rapid pressure drop near the wellbore and a shortened stable production period, which adversely affects the ultimate recovery factor. In summary, the existing well pattern and fracture network show significantly insufficient control over the reserves, resulting in poor reservoir utilization. Therefore, it is necessary to conduct a systematic well pattern–fracture network coupled optimization study. This work aims to reconstruct the well pattern and hydraulic fracture system to improve reservoir utilization and enhance overall development efficiency.

### Optimization Scheme

Accurate characterization of geological structures is essential for ensuring the precision of the subsequent well pattern–fracture network coupled optimization. To achieve this, a detailed stratigraphic framework and fault network were established based on seismic interpretation results, mud logging data, well logging curves, and core analysis from the target block. Stratigraphic correlation and fault dissection were performed

to further refine the geological structure. Given the significant reservoir heterogeneity and complex phase behavior, multi-scale geological modeling techniques were employed. A sequence of facies-controlled and lithology-controlled reservoir facies models was constructed to reveal the heterogeneous distribution patterns of the reservoir. Furthermore, the Sequential Gaussian Simulation method was applied to build a facies–lithology-controlled reservoir model and a matrix porosity–permeability model for the tight oil reservoir in Block G. As a result, a detailed three-dimensional geological model suitable for reservoir numerical simulation was developed. Subsequently, the static reservoir parameters were imported into the reservoir numerical simulation software. Combined with the fluid properties and dynamic monitoring data of the target block, a reservoir numerical simulation model was constructed. History matching was conducted by adjusting key parameters, including porosity, permeability, and relative permeability curves, to accurately reproduce the production performance. On this basis, the control variables and their value ranges for the coupled optimization were determined according to the well pattern–fracture network optimization parameter system. Field construction capabilities and engineering experience were also considered during this process. Details of the control variables are provided in [Table 1](#).

**Table 1—Optimization Parameter Settings for Well Pattern and Fracture Network.**

| Parameter Category | Parameter                     | Initial Value | Minimum | Maximum |
|--------------------|-------------------------------|---------------|---------|---------|
| Well Pattern       | Transverse Distance (m)       | 1200          | 500     | 4000    |
|                    | Longitudinal Distance (m)     | 200           | 100     | 300     |
|                    | Number of Rows                | 2             | 1       | 12      |
|                    | Number of Columns             | 0             | 0       | 12      |
|                    | Shear Angle (°)               | 0             | -90     | 90      |
|                    | Rotation Angle (°)            | 0             | -90     | 90      |
|                    | Horizontal Section Length (m) | 1000          | 500     | 2000    |
|                    | Azimuth (°)                   | 60            | 0       | 360     |
| Fracture Network   | Fracture Half-Length (m)      | 100           | 50      | 150     |
|                    | Fracture Width (m)            | 2             | 1.5     | 2.5     |
|                    | Fracture Height (m)           | 10            | 5       | 10      |
|                    | Fracture Orientation (°)      | 80            | 60      | 90      |
|                    | Number of Fractures           | 8             | 4       | 16      |

## Results and Analysis

Based on the established well pattern–fracture network coupled optimization method, key parameters were systematically optimized. [Figure 4](#) illustrates the variation trends of major evaluation indicators with the number of iterations during the optimization process. The results show that as the optimization progresses, each parameter gradually converges. The number of wells decreases from the initial 25 to a final stable range of 8 to 10 wells. The fracture-controlled reserves exhibit significant fluctuations in the early stage of optimization but gradually stabilize at approximately  $2.0 \times 10^4 \text{ m}^3$  in the later stages. The economic net present value (NPV) continuously improves throughout the optimization process, increasing from 60 million CNY to 120 million CNY. The cumulative oil production shows relatively small fluctuations during the entire process and eventually stabilizes at approximately  $3.0 \times 10^4$  to  $3.2 \times 10^4 \text{ m}^3$ .

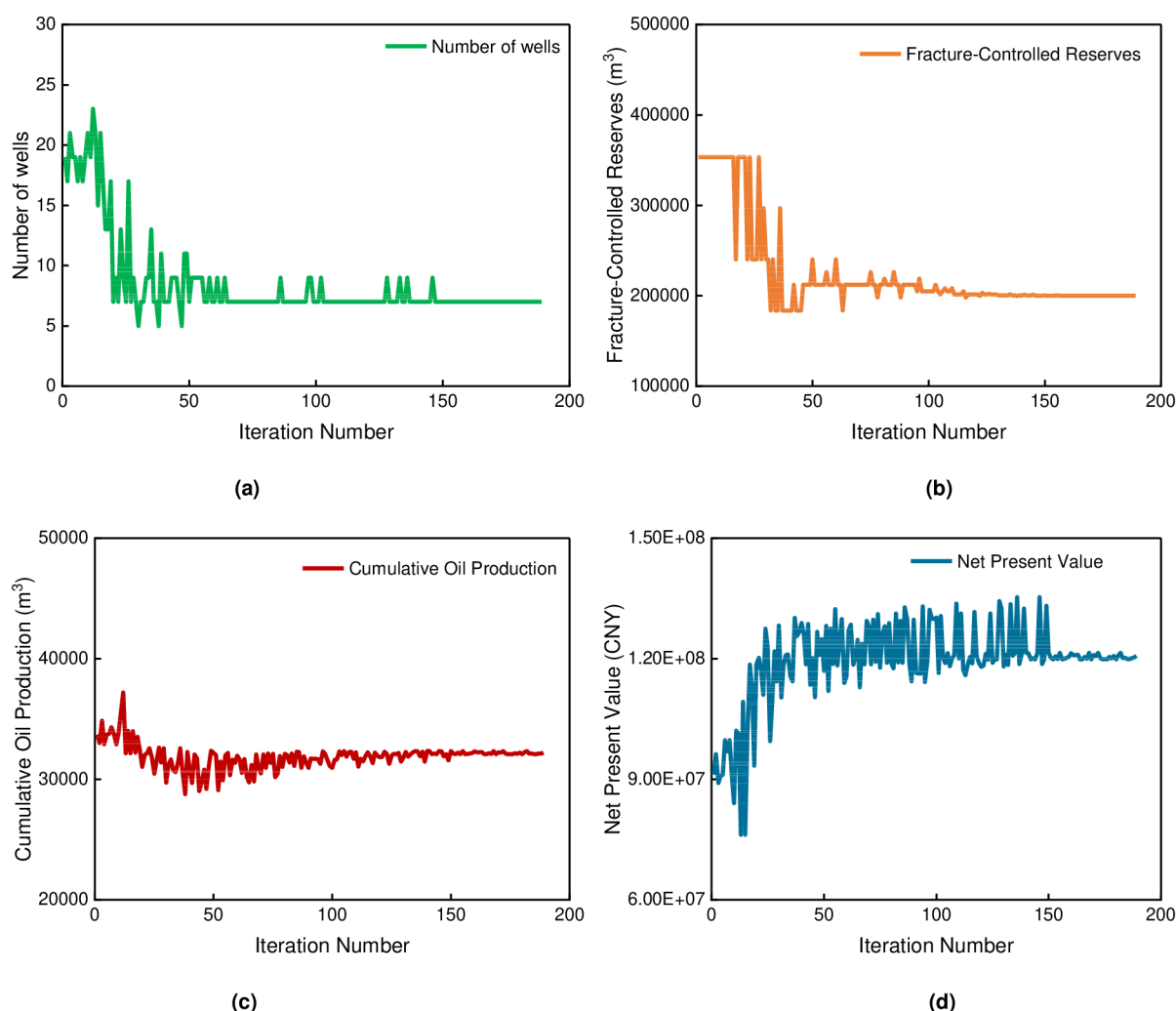


Figure 4—Trends of key evaluation indicators during the optimization process: (a) Number of wells; (b) Fracture-controlled reserves; (c) Cumulative oil production; (d) Net present value (NPV).

To further evaluate the optimization effect, a comparative analysis of the development performance before and after optimization was conducted, combined with the evolution of the oil saturation field (Figures 5 and 6). Figure 5 illustrates the oil saturation distribution under the initial pre-optimization scenario. The results indicate that the well pattern is relatively concentrated, with some wells located in low-permeability and low-saturation regions. The coverage of high oil saturation zones is insufficient, resulting in limited overall reservoir utilization. By the end of the development period, the average oil saturation remains at 0.52, and the coverage of high oil saturation zones ( $S_o > 0.5$ ) is approximately 37%. In contrast, Figure 6 shows the saturation evolution characteristics under the optimized well pattern–fracture network configuration. After optimization, the well pattern is restructured, and the number of wells increases from 4 to 7. The coverage of high oil saturation zones is significantly improved, and the fracture-controlled drainage area is expanded, enabling more effective utilization of the enriched zones. By the end of production, the average oil saturation decreases to 0.43, which represents a reduction of approximately 17.3% compared to the pre-optimization scenario. Meanwhile, the coverage of high oil saturation zones increases to 81%. The area of residual oil accumulation is significantly reduced, indicating a substantial improvement in reservoir utilization.

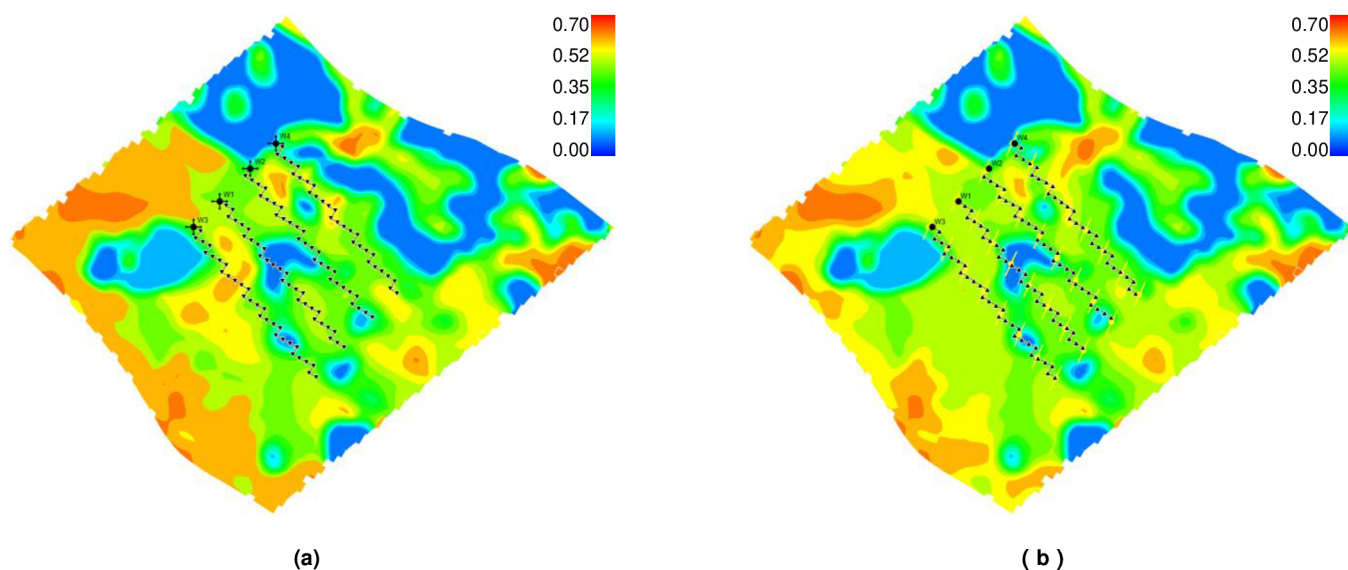


Figure 5—Oil saturation distribution under the initial well pattern–fracture network configuration: (a) Initial oil saturation; (b) Remaining oil saturation after development.

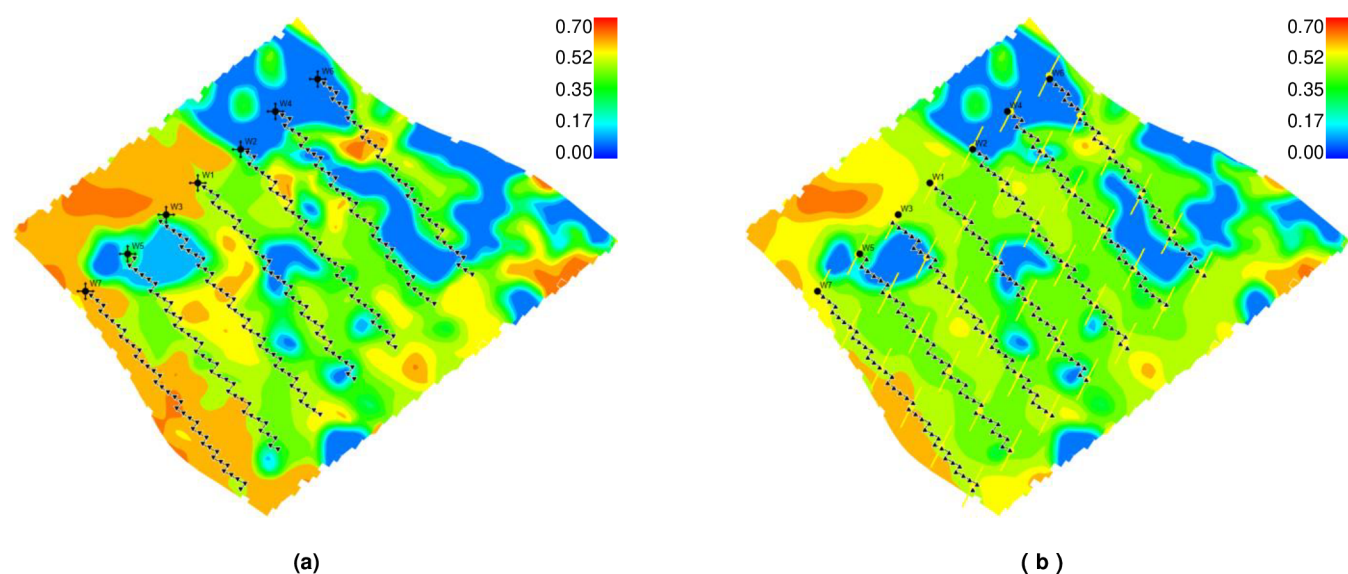


Figure 6—Oil saturation distribution under the optimized well pattern–fracture network configuration: (a) Initial oil saturation; (b) Remaining oil saturation after development.

The trends of the key evaluation indicators (Figure 4), together with the evolution characteristics of the saturation field (Figures 5 and 6), jointly verify the effectiveness of the well pattern–fracture network coupled optimization method. This method overcomes the limitations of traditional stepwise optimization and achieves a global coupled optimization of development scale, reservoir utilization, and economic performance. In the early stage of the optimization process, the focus is primarily on adjusting the well pattern configuration. By progressively eliminating redundant wells, the development cost is effectively reduced. In the subsequent stage, the optimization shifts toward fine-tuning fracture parameter configurations, which continuously enhances the fracture-controlled reserves and reservoir utilization. The steady increase in NPV indicates that the optimization path effectively balances development investment and production revenue, demonstrating both strong physical rationality and practical engineering applicability. Meanwhile, the SADE–Sammon optimization algorithm employed in this study successfully integrates global exploration and local search mechanisms. It demonstrates excellent solution space search capability



and superior convergence performance, ensuring a clear convergence path and highly stable and reliable optimization results.

### Field Application and Validation

To verify the engineering applicability of the proposed well pattern–fracture network coupled optimization method, a field application test was conducted in a typical tight oil reservoir in Block G. Before the implementation of the optimized scheme, the average daily liquid production per well across the block was approximately 3.5 m<sup>3</sup>/d.

Based on the results of the proposed optimization model, targeted measures were applied to the existing well pattern. These included well shut-ins, perforation adjustments, and refracturing of the old wells. Additionally, new wells were drilled and hydraulically fractured according to development requirements. After two years of development and production, the average daily liquid production per well in the first year increased significantly to 11.6 m<sup>3</sup>/d. Although the overall development scale remained largely unchanged, the fracture-controlled reserves increased by 20.41% compared to the pre-optimization scenario, and the effective drainage area of the reservoir was significantly expanded. Production forecasts indicate that the 10-year cumulative oil production per well under the optimized scheme is approximately 9000 m<sup>3</sup> higher than that of the original scheme, demonstrating simultaneous improvement in both reservoir utilization and economic performance. In terms of computational efficiency, compared to traditional direct optimization methods based on numerical simulator kernels (such as genetic algorithms or differential evolution), the introduction of the SADE–Sammon algorithm reduced the total optimization computation time by approximately 43.14%. This significantly improved optimization efficiency while maintaining solution accuracy.

In summary, the well pattern–fracture network coupled optimization method developed in this study has been successfully validated through field application. The method demonstrates substantial advantages in enhancing reservoir utilization, improving development efficiency, and reducing computational time, indicating strong potential for broader engineering application.

### Conclusions

To address the challenge of well pattern–fracture network coupled optimization in tight oil reservoir development, this study proposed a coupled optimization method that integrates geometric parameterization, multi-objective optimization, and intelligent algorithmic solutions. Field application and validation were conducted in a typical tight oil reservoir in Block G of the Bohai Bay Basin. The main findings and conclusions are summarized as follows:

1. A dynamically evolving well pattern unit model was developed based on spatial topology analysis and geometric transformation theory, incorporating spatial safety constraints. This model enables the quantitative characterization of the geometric configuration of the well pattern–fracture network. The proposed method effectively overcomes the weak parameter coupling and the difficulty of embedding safety constraints in traditional modeling, providing a high-accuracy physical foundation for coupled optimization.
2. To address multi-objective development requirements, a multi-constraint nonlinear optimization model was established. The core objectives of this model are to maximize fracture-controlled reserves and achieve balanced and effective reservoir utilization. The calculation of fracture-controlled reserves introduces a graded weighting mechanism based on stimulation coefficients, which significantly improves the accuracy of reservoir utilization evaluation. Meanwhile, the NPV model dynamically links well pattern scale, fracture parameters, and development costs, enabling closed-loop optimization of both economic performance and reservoir development efficiency.



3. In this study, a nested iterative optimization framework was constructed, driven by Latin Hypercube Sampling, Kriging surrogate modeling, and dimensionality reduction mapping. This framework simultaneously achieves global exploration and local fine-tuning within a limited number of reservoir simulation evaluations. Compared with traditional direct optimization algorithms (such as genetic algorithms or differential evolution), the proposed method significantly improves optimization efficiency and reduces computation time by approximately 43.14%. This provides an effective solution for high-dimensional, strongly coupled optimization problems.
4. Following the implementation of the optimized scheme in the Block G tight oil reservoir, the average daily liquid production per well in the first year increased from 3.5 m<sup>3</sup>/d to 11.6 m<sup>3</sup>/d. The fracture-controlled reserves increased by 20.41%, and the predicted 10-year cumulative oil production per well increased by 9000 m<sup>3</sup>. These results demonstrate simultaneous improvement in both reservoir utilization and economic performance, verifying the engineering applicability and practical value of the proposed method.
5. Overall, the proposed method successfully overcomes the limitations of traditional stepwise optimization, including insufficient parameter coverage and single-objective focus. It provides a new technical pathway for the efficient development of tight oil reservoirs. Future work will further integrate production dynamic data into the optimization process to develop full life-cycle dynamic optimization technology.

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